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Unit 1 - Topic Investigation

PDF Documents

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This PDF document is included in the unit materials.

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UNIT 1 TOPIC INVESTIGATION

Introduction

In this first step, your team will work together to choose a project topic that everyone is excited about. You'll learn how to brainstorm ideas, research your topic, and make sure it's something your team can realistically achieve. This is an important first step because a good project starts with a great idea!



Here's what you'll do:

BRAINSTORMING TOGETHER

Share ideas with your teammates and discuss which ones you're most interested in. You'll work together to pick the best idea for your project.

RESEARCHING YOUR TOPIC

After you choose your project topic, you'll dig deeper into it. Each team member will do some research and share what they've learned with the group.

FINALIZING THE TOPIC

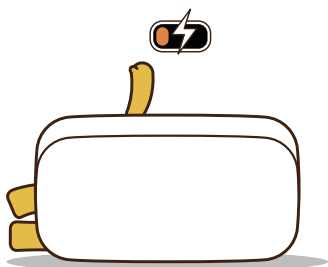
Once your research is done, you'll decide on the final project topic together, making sure it's something your team can complete successfully.

UNDERSTANDING TEAM ROLES

You'll assign roles within your team, making sure everyone knows what they're responsible for. This helps the project run smoothly and ensures that everyone contributes.

SETTING GOALS

Finally, you'll set some goals and milestones for your project. This will help your team stay on track and make sure you finish on time.



Do you remember your self project? For your self project, you did everything yourself. However, this team project requires you to collaborate with your team members from brainstorming to the final report.



Creations by Students Through Team Projects



- **MODI Modules:**
Network, Display, Motor A, Dial, Button, Speaker
- **Additional Materials:**
Paper boxes, glue, sticky notes, etc.
- **How It Works:**
Use the dial to choose one of three different beverages. The display shows your choice. Then, if you press the button, Motor A moves to a specific position to pour the chosen beverage. The speaker plays a sound while the beverage is pouring and when it's done.

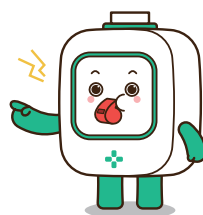
The Smart Beverage Vending Machine project involved students working together to design and build a device that dispenses one of three beverages. Using MODI modules like the Dial, Display, Motor A, Button, and Speaker, along with paper boxes for the structure, the students programmed the machine to allow beverage selection and dispensing with sound effects. The project required teamwork in planning, building, programming, and testing to ensure smooth operation.

Unit Overview

In Unit 1, your team will explore and identify a suitable topic for your self project. This unit will guide you through the following processes:

1. IDENTIFYING NEEDS FOR A SMART DEVICE

- Think about various tasks or activities that could be improved with a smart device. Consider different environments or communities that could benefit from your project.



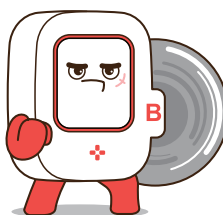
2. RESEARCHING EXISTING SMART DEVICES

- Investigate existing smart devices for inspiration. Note what problems they solve, how they work, and any useful features that could be applied to your project.



3. CHOOSING YOUR PROJECT TOPIC

- Select a specific topic for your team project based on your research. Consider the feasibility, impact, and your team's interest in the idea.



4. QUESTIONING THE DETAILS

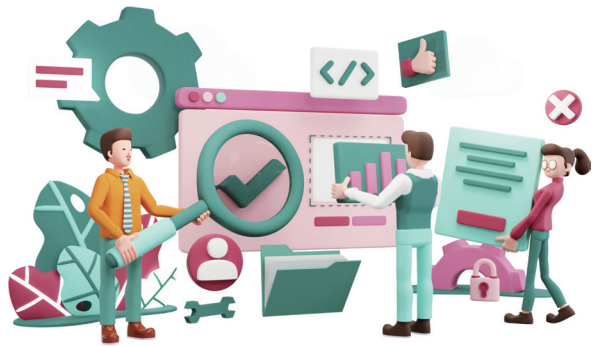
- Answer key questions about the tasks to be improved, necessary features, existing devices for inspiration, and the resources required for your project.



Identifying Needs for a Smart Device

Working Together to Identify Needs

In this step, your team will brainstorm and discuss different areas where a smart device could make a difference. This could be in your school, community, or even in public spaces. The goal is to identify a problem or a task that can be improved with a smart device.



HOW TO DO IT:

1. Brainstorm as a Team:

- Each team member should share ideas about tasks or activities that could be improved with a smart device.
- Think broadly—consider areas like safety, convenience, efficiency, or environmental impact.

2. Discuss and Refine Ideas:

- After everyone has shared their ideas, discuss them together.
- Identify common themes or problems that interest the entire team.
- Narrow down the list to a few key needs that your project could address.

3. Agree on a Focus:

- As a team, decide on the most promising area to focus on.
- Make sure everyone is excited about and committed to the chosen focus.

TIPS

Encourage every team member to participate in the discussion. Great ideas often come from different perspectives!



● Researching Existing Smart Devices

Learning from What's Already Out There

Now that your team has identified a need, it's time to look at how others have addressed similar problems. By researching existing smart devices, you can get inspiration and avoid potential pitfalls.



HOW TO DO IT:

1. Divide the Research:

- Assign different team members to research various types of smart devices that could inspire your project.
- Look for devices that solve similar problems or have features that might be useful.

2. Gather Information:

- Each team member should find out how these devices work, what problems they solve, and what features they include.
- Pay attention to what works well and what could be improved.

3. Share and Discuss:

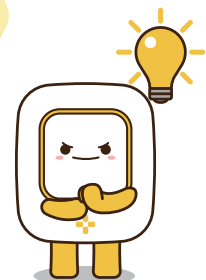
- Meet as a team to share what each person has learned.
- Discuss how the features of these existing devices could be applied to your project.

4. Document Your Findings:

- Keep detailed notes on the devices you researched.
- This information will help you design your own smart device.

TIPS

Make sure everyone shares their findings clearly so that the whole team benefits from the research.





Choosing Your Project Topic

Deciding on the Best Idea Together

With all the research and brainstorming done, it's time for your team to choose a specific project topic. This step is crucial because it sets the direction for all your future work.



HOW TO DO IT:

1. Review Your Options:

- Look at the different needs and ideas your team has identified.
- Consider how feasible each idea is—think about the resources you have and the skills needed.

2. Discuss Feasibility:

- As a team, discuss which ideas are most practical and impactful.
- Consider the time you have, the materials you'll need, and the complexity of the project.

3. Make a Decision:

- Vote or come to a consensus on the project topic.
- Make sure everyone feels good about the choice and is ready to commit to it.

4. Document Your Decision:

- Write down your chosen topic and why you selected it.
- This will help keep your team focused and on the same page.

TIPS

Ensure that everyone's voice is heard in the decision-making process. The best decisions are made when everyone's opinions are considered.



Questioning the Details

Key Questions to Guide Your Project

Now that your team has chosen a project topic, it's time to think through the details. Answering these key questions will help you clarify your project's purpose and plan:

What task will your smart device improve?

- Clearly define the problem your device will solve.



What features are necessary for your device to work effectively?

- Identify the must-have features that will make your device successful.



What materials and resources will you need?

- List the tools, materials, and MODI modules required to build your device.





Conclusion

Reflecting on What We've Accomplished

Congratulations! Your team has successfully completed the first unit of your project. You've worked together to brainstorm ideas, conduct research, and choose a project topic that everyone is excited about. By answering key questions and planning out the details, you've laid a strong foundation for the rest of your project.



WHAT'S NEXT?

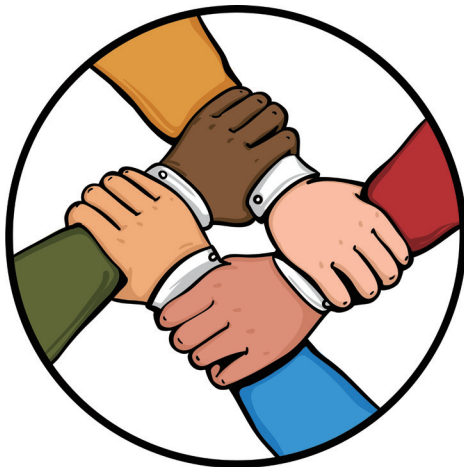
In the next unit, your team will start bringing your ideas to life. You'll begin designing your smart device, selecting the right MODI modules, and planning how it will all come together. The work you've done in this unit will guide you through the next steps, ensuring that your project is well-planned and ready to move forward.



Teamwork Reminder

Remember, the success of your project depends on how well you work together. Keep communicating, supporting each other, and making sure everyone's voice is heard. Great teamwork will lead to a great project!

The Importance of Teamwork in the 21st Century



In today's world, the ability to work well with others is more important than ever.

The 21st century is marked by rapid technological advancements, globalization, and complex challenges that no one person can solve alone.

Here's why teamwork is essential:

SOLVING PROBLEMS TOGETHER

Big challenges need different ideas and skills. Working as a team helps us find better solutions.

CREATIVITY & NEW IDEAS

When people work together, they come up with creative ideas that one person might not think of alone.

LEARNING IMPORTANT SKILLS

Teamwork helps you practice communication, leadership, and cooperation—skills you'll need in the future.

WORKING WITH PEOPLE EVERYWHERE

In our connected world, we often need to work with people from different places. Teamwork prepares you for that.

PREPARING FOR THE FUTURE

Being a good team player will help you in school, in your job, and in life. It's one of the most valuable skills you can have!

By working together on this project, you're building these important skills that will help you succeed in the future. Keep up the great teamwork!

